Diego Arteaga

(661) 219-4382 | darteaga520@gmail.com | darteaga.dev | linkedin.com/in/arteagadiego

Education

California State University, Northridge — Northridge, CA
Master of Science in Data Science — *May 2027*
Bachelor of Science in Computer Science (Summa Cum Laude) — *May 2025*

- **Relevant Coursework in:** Data Structures, Algorithm Design, Software Engineering, SQL Database Design, Big Data, Python for Data Science & Analytics, Data Visualization, Machine Learning, Deep Learning, Probability & Statistics

Experience

Outreach Representative — Sept 2025 - Present
Los Angeles Valley College — *Los Angeles, CA*

- Cleaned, maintained, and updated confidential student records using PeopleSoft and **Microsoft Excel**
- Managed a caseload of 100+ high school students while performing data validation, record reconciliation, and reporting to support outreach, recruitment, and leadership operations and maintain database integrity
- Supported college fairs, registration, and commencement events, engaging with prospective students and parents

Social Media Manager & Content Creator — Nov 2021 - Present
YouTube (Self-Employed) — *Los Angeles, CA*

- Analyzed content performance and managed YouTube Shorts, TikTok, and Instagram for a 350k+ subscriber creator brand (Arrowstotle), optimizing publishing strategy and audience engagement
- Built **Google Sheets** dashboards to track KPIs, monitor trends, and support data-driven content strategy decisions
- Independently managed personal pages and conducted **A/B tests** on content formats and thumbnail designs across monetized personal channels, increasing revenue by **22%** and engagement by **40%**

Undergraduate Research Intern — June 2025 - Aug 2025
California State University, Northridge — *Northridge, CA*

- Developed components of a deep learning-based anomaly detection model for healthcare IoT traffic data using **Python, PyTorch,** and **Scikit-learn** in **Google Colab**; preprocessed and analyzed large-scale IoT network data
- First author of an **IEEE-published research paper** and presented findings at the IEEE IEMCON 2025 conference

Projects

Healthcare IoT Network Anomaly Detection | *Python, PyTorch, Scikit-learn, Google Colab*

- Built an anomaly detection pipeline using Transformer-based architectures and temporal-graph feature engineering to identify malicious behavior in healthcare IoT networks
- Project recognition: **Best Paper Presentation, IEEE IEMCON 2025**

Japanese-to-English Fusion Translation Model | *Python, PyTorch, Pandas, Linux, Transformers, Hugging Face, NLP*

- Processed a 3-million-sentence bilingual corpus and conducted ablation studies of novel fusion methods, evaluating Transformer-based machine translation models using BLEU and chrF in a **Linux GPU environment**
- Developed metadata-guided and BERTScore-based fusion methods to improve domain-specific translation quality

MLB Team-Season Success Analytics | *Python, Pandas, Matplotlib, PyMC*

- Processed and analyzed historical Lahman Baseball data using Pandas, performing data cleaning, joins, aggregations, and feature engineering to evaluate team performance across MLB seasons
- Analyzed batting, pitching, and fielding factors to model future MLB team performance using PyMC

Agricultural IoT & Machine Learning Platform | *React, Bootstrap, Node.js, Python, Postman, Firebase*

- Designed a real-time dashboard using React and Bootstrap to visualize environmental sensor data from Firebase within a team of 12 students via GitHub and Jira, integrating and testing **Python backend APIs**
- Collaborated on machine learning models to correlate environmental data with plant sickness and crop growth
- Project recognition: **1st Place, CSUNposium 2025**

Skills

Programming Languages: Python, SQL, Java, JavaScript, HTML/CSS
Technologies: NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch, MySQL, Google BigQuery, Git, GitHub, GitHub CoPilot, React, Node.js, Express.js, Jupyter Notebook, Google Colab, VS Code, AWS, Jira, NPM
Languages: English, Spanish, Japanese